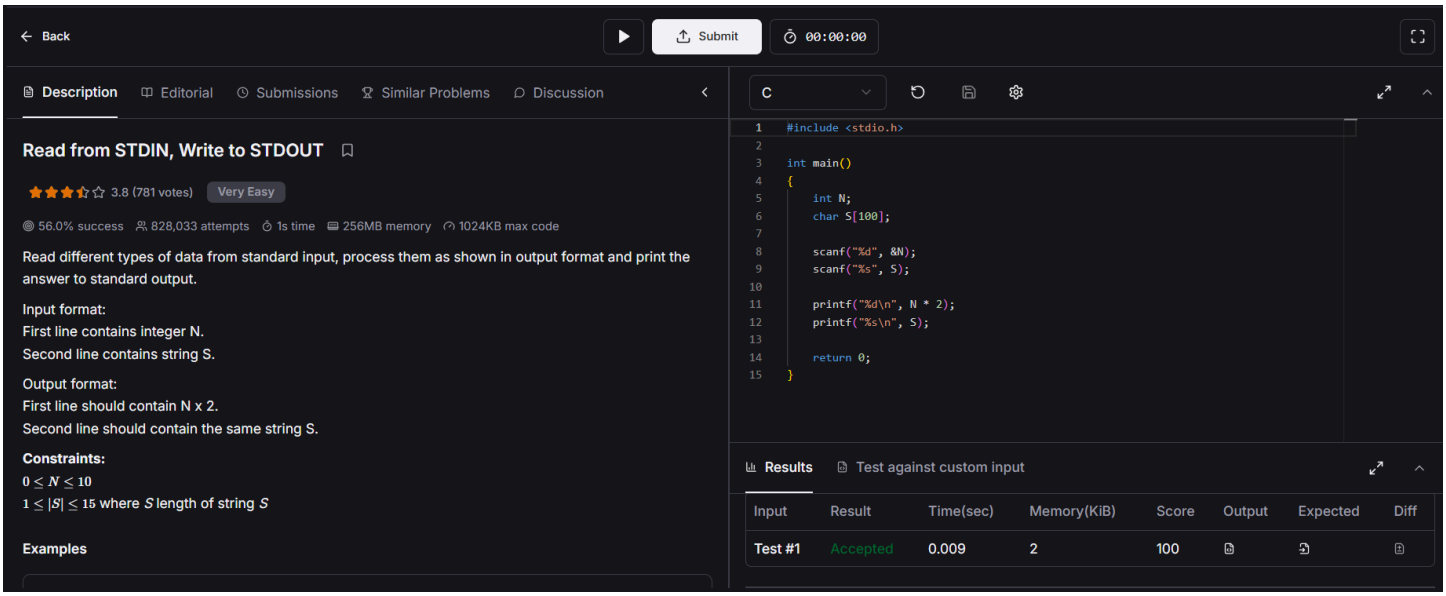
C Programming Platform Coding Practice Question

Module 1 — Input/output, variables and basic arithmetic

QUESTION–1 [Read from STDIN, Write to STDOUT]

[Problem Statement Link]= [https://www.hackerearth.com/practice/basic-programming/input-output/basics-of-input-output/practice-problems/algorithm/io-exercise/]



[Source Code]

```c
#include <stdio.h>

int main()
{
    int N;
    char S[100];

    scanf("%d", &N);
    scanf("%s", S);

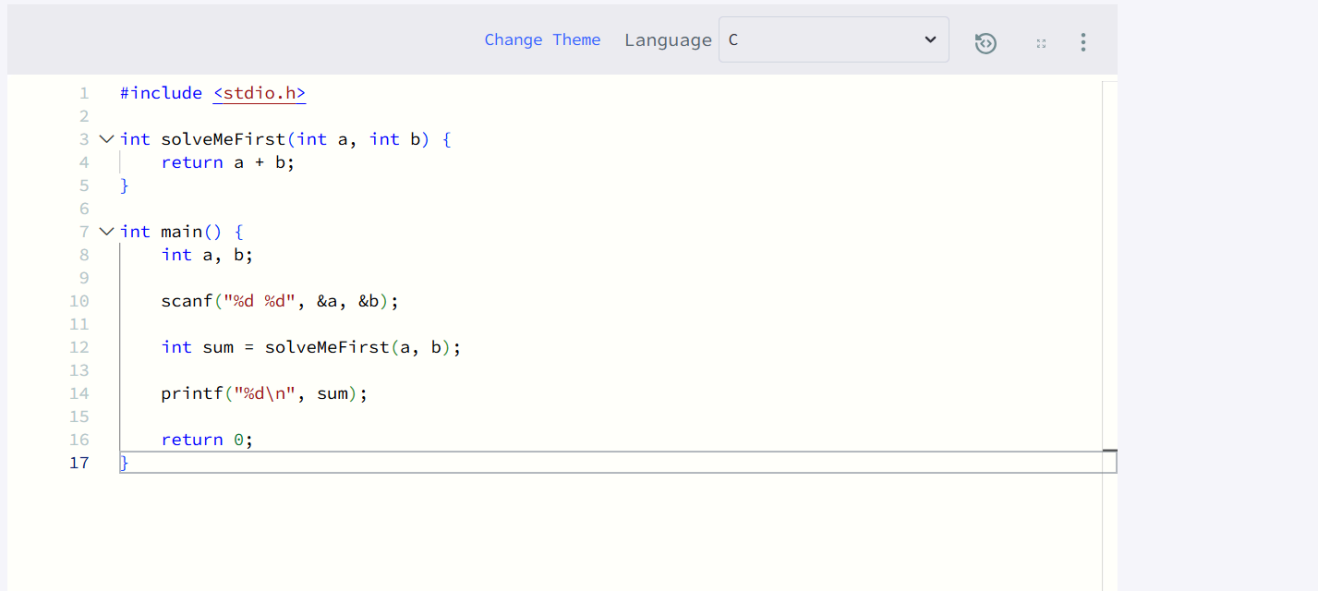
    printf("%d\n", N * 2);
    printf("%s\n", S);

    return 0;
}
```

QUESTION-2

[Problem Statement Link]=[Solve Me First | HackerRank]

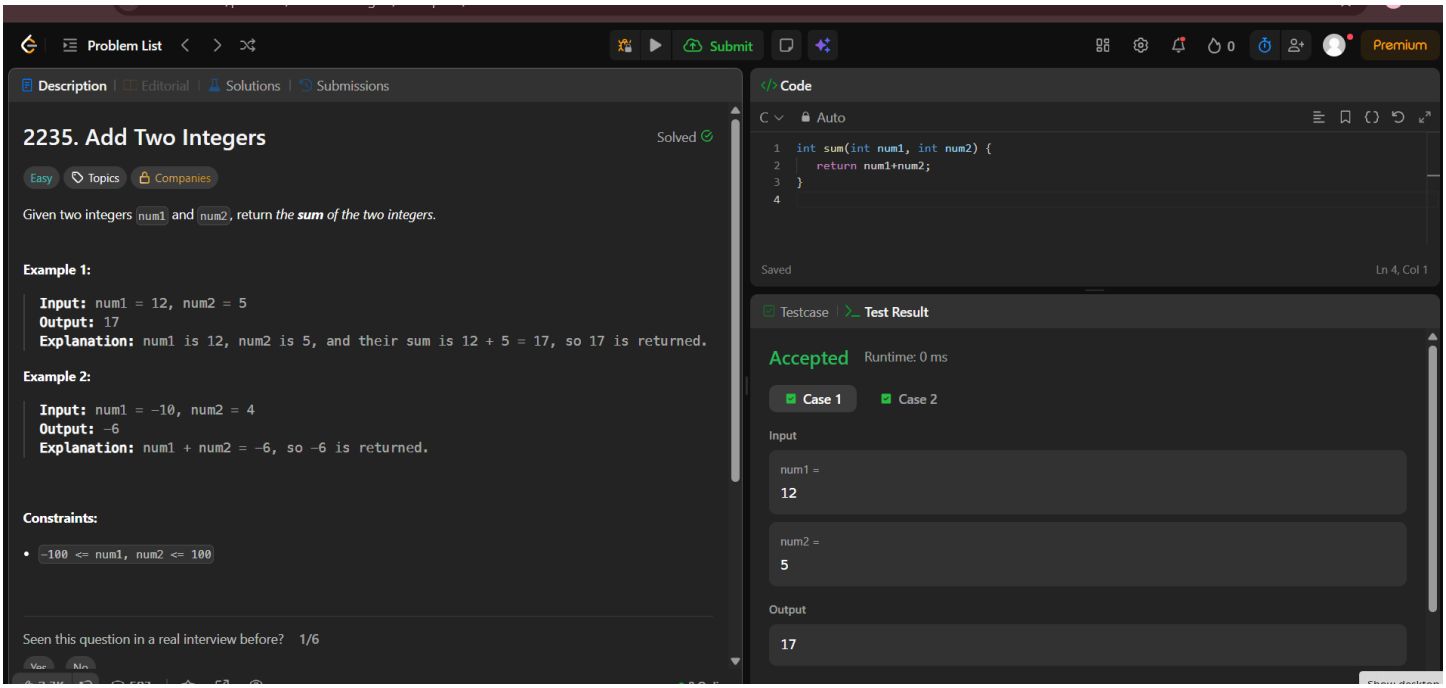
[Source Code]



QUESTION-3

[Problem Statement Link]=[Add Two Integers - LeetCode]

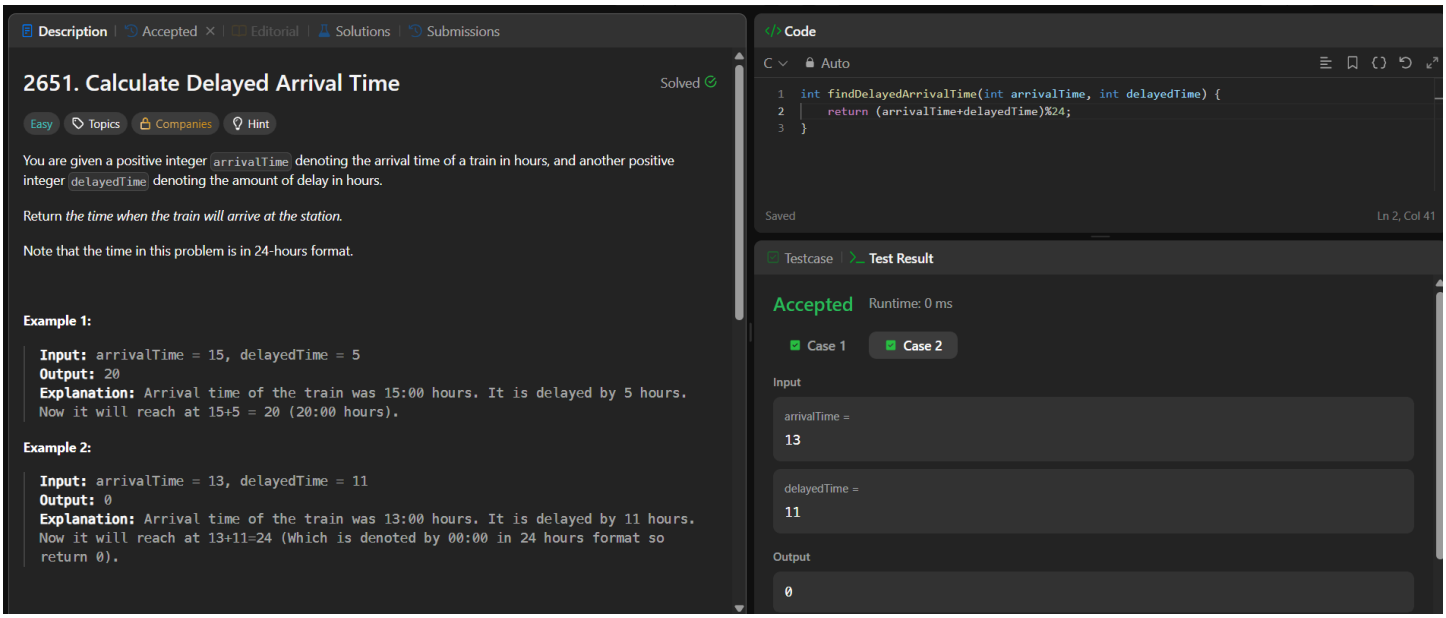
[Source Code]



QUESTION-4

[Problem Statement Link]=[Calculate Delayed Arrival Time - LeetCode]

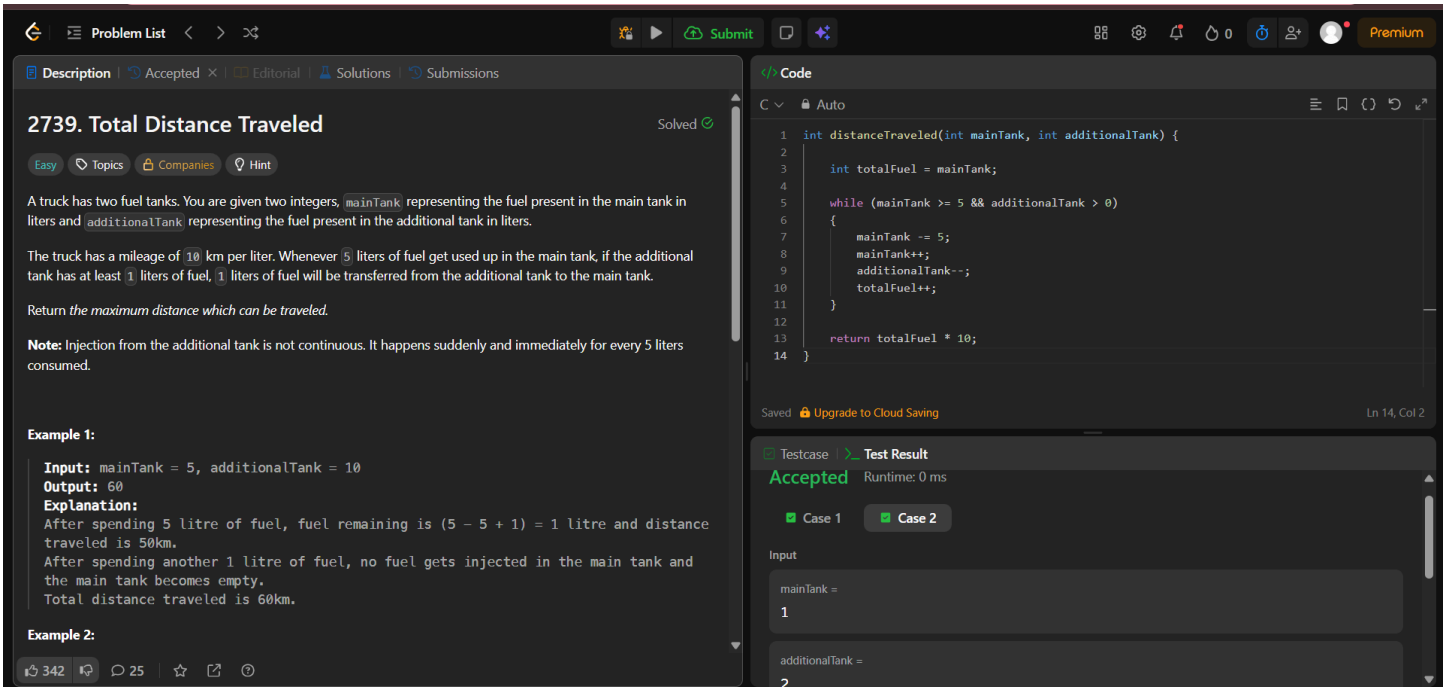
[Source Code]



QUESTION–5

[Problem Statement Link]=[Total Distance Traveled - LeetCode]

[Source Code]



Module 2 — Conditional statements and operators

QUESTION–1

[Problem Statement Link]=[Conditional Statements in C | HackerRank]

[Source Code]

Change Theme Language: C

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    if (n == 1) {
        printf("one");
    }
    else if (n == 2) {
        printf("two");
    }
    else if (n == 3) {
        printf("three");
    }
    else if (n == 4) {
        printf("four");
    }
    else if (n == 5) {
        printf("five");
    }
    else if (n == 6) {
        printf("six");
    }
    else if (n == 7) {
```

Line: 1 Col: 1

QUESTION-2

[Problem Statement Link]=[Grading Students | HackerRank]

[Source Code]

Change Theme Language C

```c
#include <stdio.h>

int main() {
    int n, grade;
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &grade);

        if (grade < 38) {
            printf("%d\n", grade);
        }
        else if (grade % 5 >= 3) {
            grade = grade + (5 - grade % 5);
            printf("%d\n", grade);
        }
        else {
            printf("%d\n", grade);
        }
    }

    return 0;
}
```

QUESTION-3

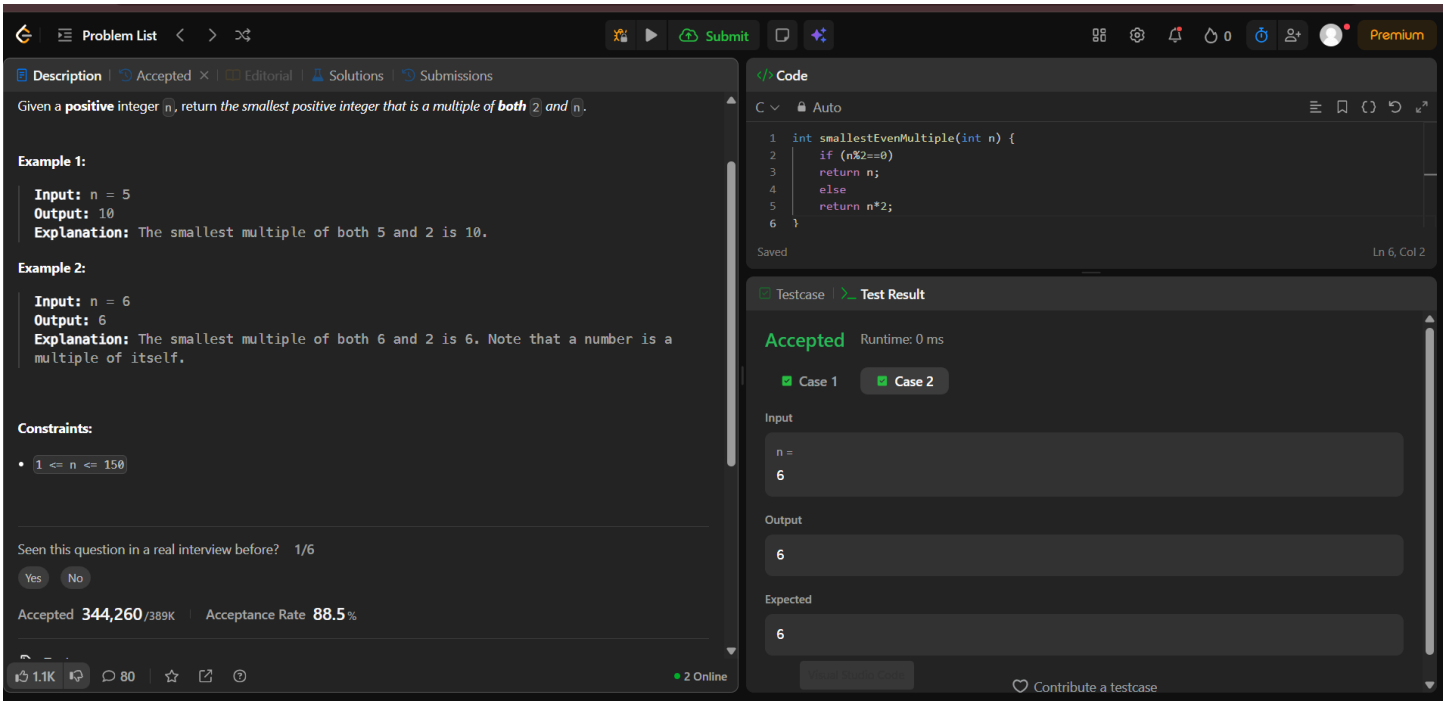
[Problem Statement Link]=[Smallest Even Multiple - LeetCode]

[Source Code]

Code

C Auto

```c
int smallestEvenMultiple(int n) {
    if (n%2==0)
    return n;
    else
    return n*2;
}
```



QUESTION–4

[Problem Statement Link]=[
🌐 Roy and Profile Picture | Basics of Input/Output & Basic Programming Practice Problems]

[Source Code]

C

```c
#include <stdio.h>

int main() {
    int L, N;
    int i;

    scanf("%d", &L);
    scanf("%d", &N);

    for (i = 0; i < N; i++) {
        int W, H;
        scanf("%d %d", &W, &H);

        if (W < L || H < L) {
            printf("UPLOAD ANOTHER\n");
        }
        else if (W == H) {
            printf("ACCEPTED\n");
        }
        else {
            printf("CROP IT\n");
        }
    }

    return 0;
}
```

QUESTION–5

[Problem Statement Link]=[🌐 FizzBuzz | HackerRank]

[Source Code]

Change Theme Language C

```c
#include <stdio.h>

int main() {
    int i;

    for (i = 1; i <= 100; i++) {
        if (i % 3 == 0 && i % 5 == 0) {
            printf("FizzBuzz\n");
        }
        else if (i % 3 == 0) {
            printf("Fizz\n");
        }
        else if (i % 5 == 0) {
            printf("Buzz\n");
        }
        else {
            printf("%d\n", i);
        }
    }

    return 0;
}
```

Module 3 — Loops and repeated computation

QUESTION-1

[Problem Statement Link]=[For Loop in C | HackerRank]

[Source Code]

Change Theme Language: C

```c
#include <stdio.h>

int main() {
    int a, b;
    scanf("%d %d", &a, &b);

    for (int i = a; i <= b; i++) {
        if (i == 1) {
            printf("one\n");
        }
        else if (i == 2) {
            printf("two\n");
        }
        else if (i == 3) {
            printf("three\n");
        }
        else if (i == 4) {
            printf("four\n");
        }
        else if (i == 5) {
            printf("five\n");
        }
        else if (i == 6) {
            printf("six\n");
        }
        else if (i == 7) {
            printf("seven\n");
```

Line: 39 Col: 29

Upload Code as File Test against custom input Run Code Submit Code

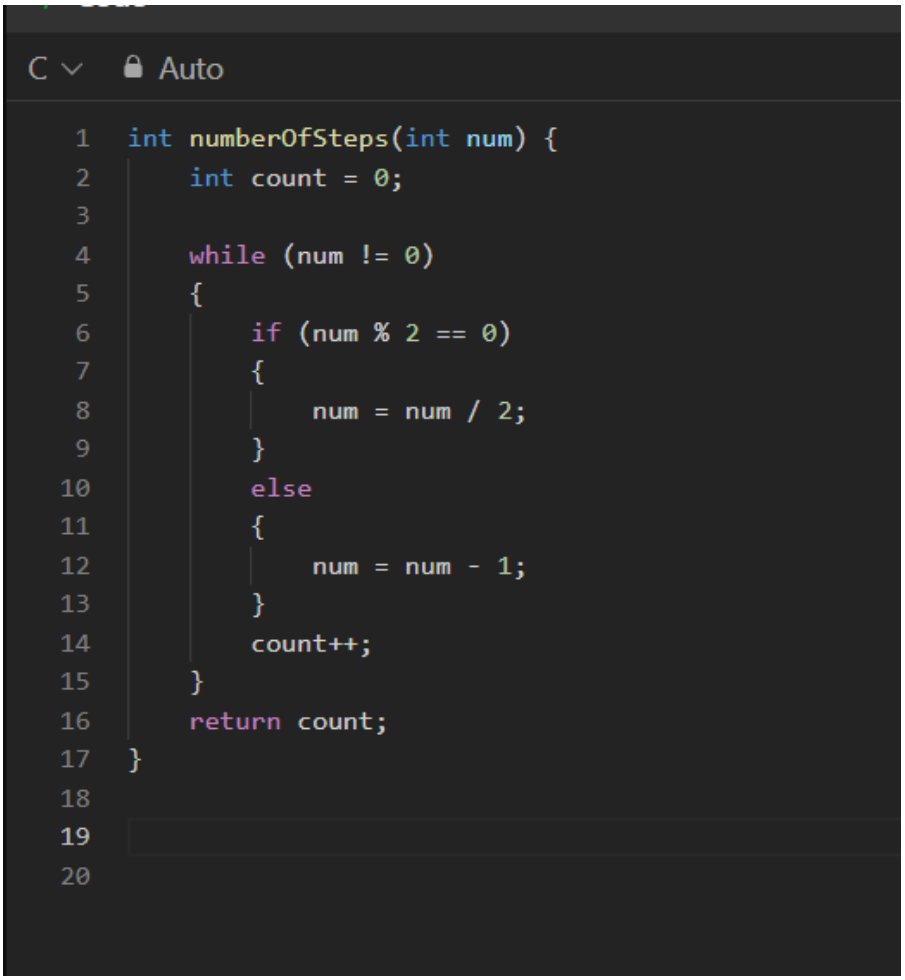
```c
int main() {
        }
        else if (i == 5) {
            printf("five\n");
        }
        else if (i == 6) {
            printf("six\n");
        }
        else if (i == 7) {
            printf("seven\n");
        }
        else if (i == 8) {
            printf("eight\n");
        }
        else if (i == 9) {
            printf("nine\n");
        }
        else if (i % 2 == 0) {
            printf("even\n");
        }
        else {
            printf("odd\n");
        }
    }

    return 0;
}
```

QUESTION-2

[Problem Statement Link]=[Number of Steps to Reduce a Number to Zero - LeetCode]

[Source Code]



QUESTION-3

[Problem Statement Link]=[Count Operations to Obtain Zero - LeetCode]

[Source Code]

Code

C Auto

```c
int countOperations(int num1, int num2) {

    int count = 0;

    while (num1 != 0 && num2 != 0) {
        if (num1 >= num2) {
            num1 = num1 - num2;
        }
        else {
            num2 = num2 - num1;
        }
        count++;
    }

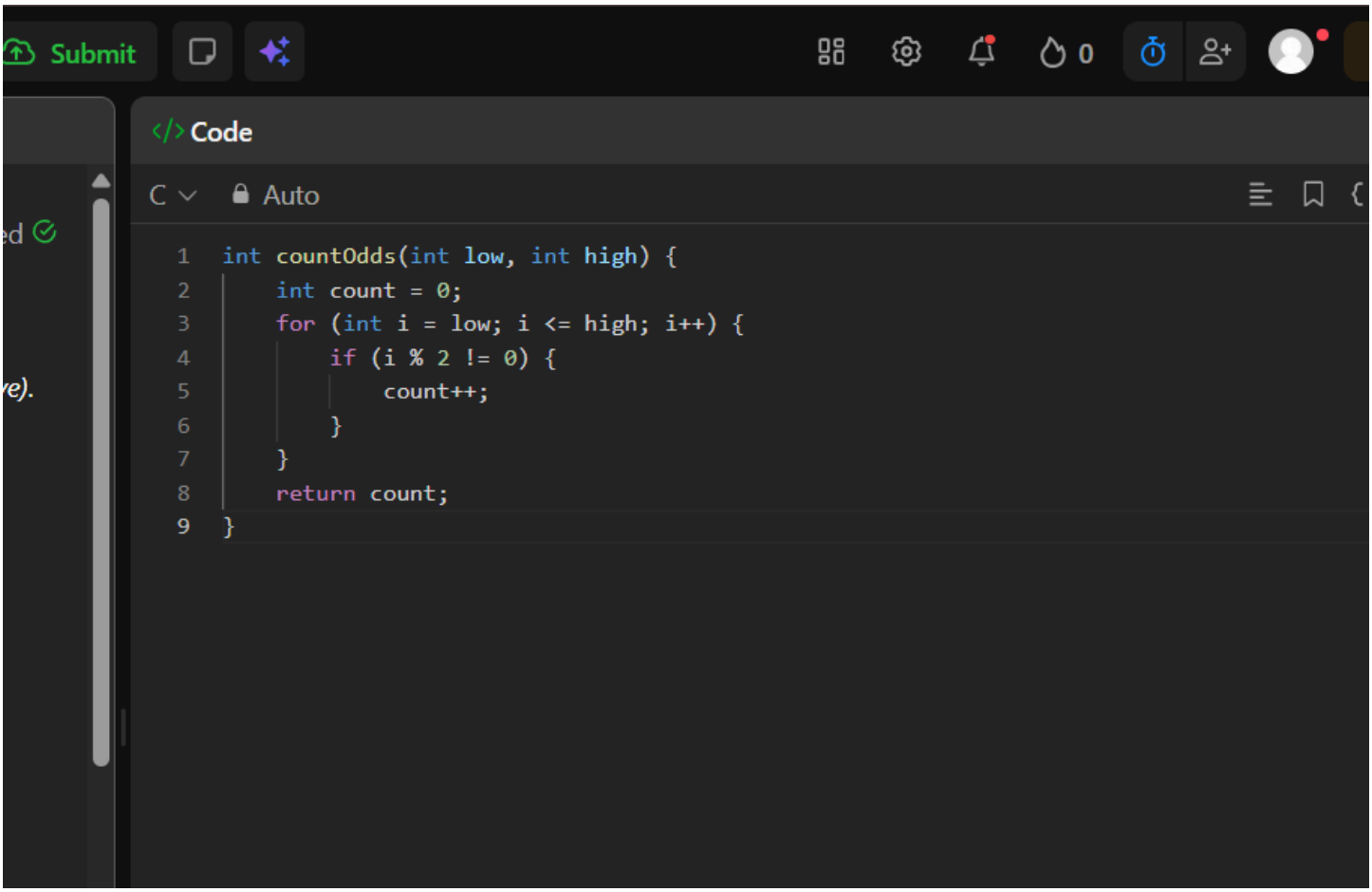
    return count;
}

```

QUESTION-4

[Problem Statement Link]=[Count Odd Numbers in an Interval Range - LeetCode]

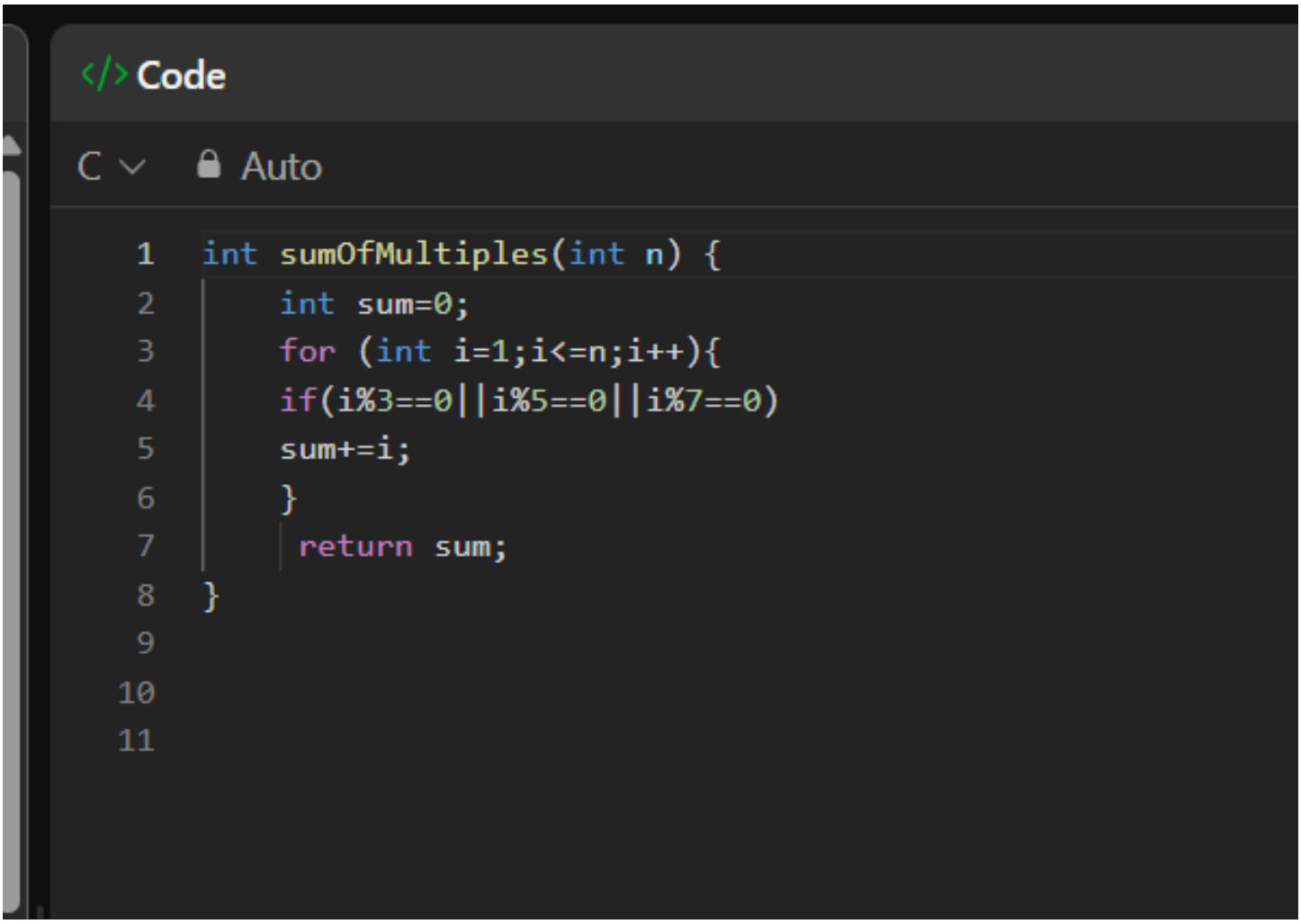
[Source Code]



QUESTION–5

[Problem Statement Link]=[Sum Multiples - LeetCode]

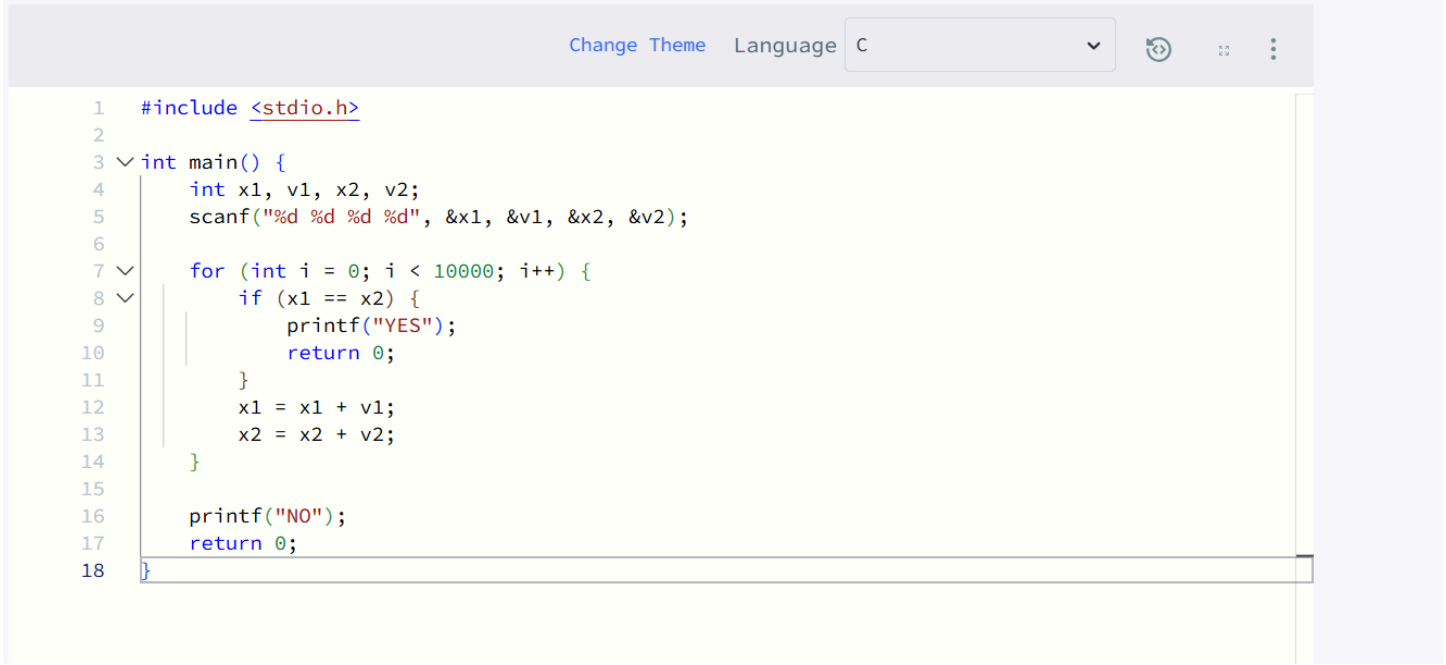
[Source Code]



QUESTION-6

[Problem Statement Link]=[Number Line Jumps | HackerRank]

[Source Code]

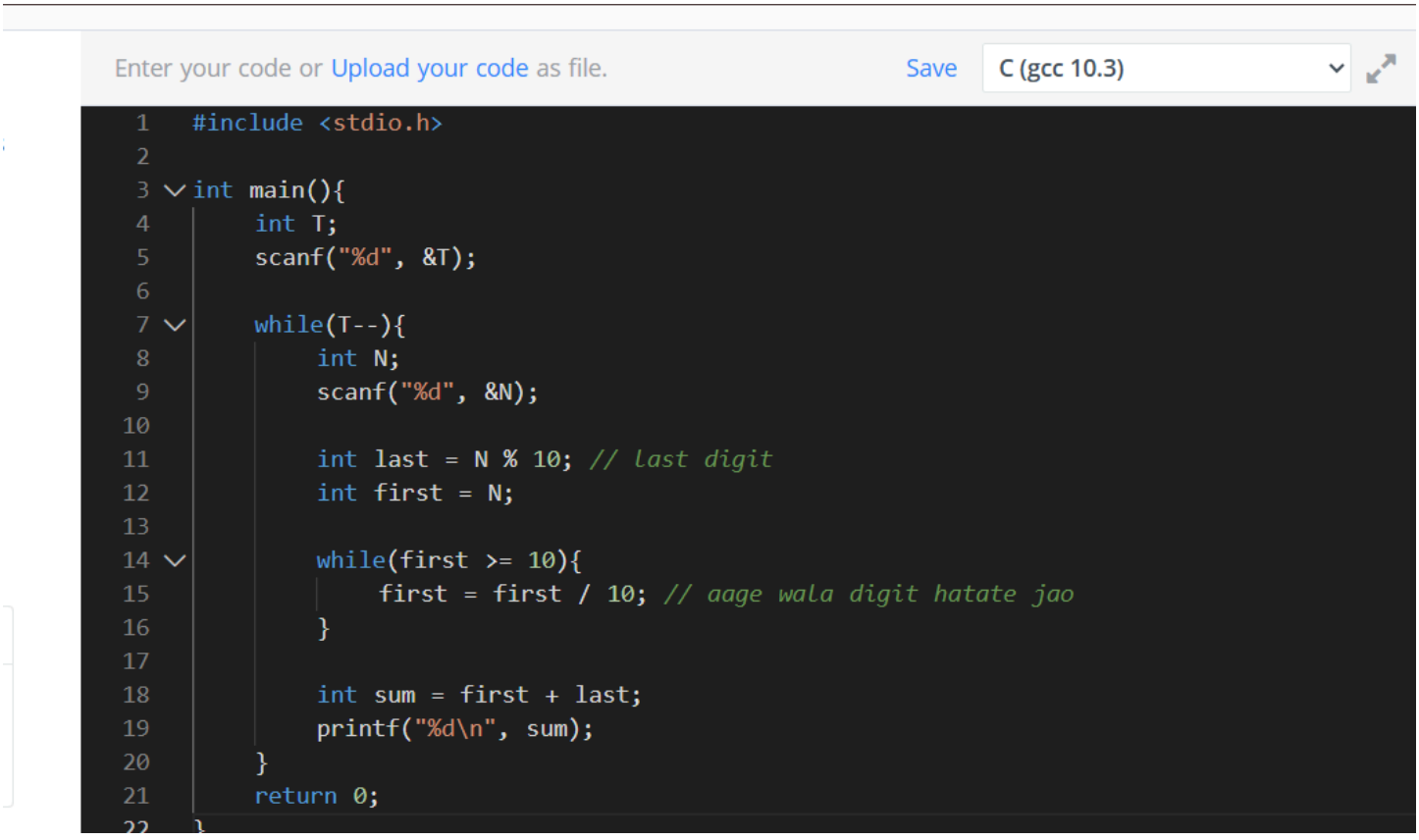


Module 4 — Digit extraction and number manipulation

QUESTION–1

[Problem Statement Link]=[A.Digit Sum | Practice Problems]

[Source Code]




```c
#include <stdio.h>

int main(){
    int T;
    scanf("%d", &T);

    while(T--){
        int N;
        scanf("%d", &N);

        int last = N % 10; // last digit
        int first = N;

        while(first >= 10){
            first = first / 10; // aage wala digit hatate jao
        }

        int sum = first + last;
        printf("%d\n", sum);
    }
    return 0;
}
```

QUESTION–2

[Problem Statement Link]=[Find Digits | HackerRank]

[Source Code]

Change Theme Language C

```c
#include <stdio.h>

int findDigits(int n) {
    int temp = n;
    int count = 0;

    while (temp > 0) {
        int digit = temp % 10;

        if (digit != 0 && n % digit == 0) {
            count = count + 1;
        }

        temp = temp / 10;
    }

    return count;
}

int main() {
    int t;
    scanf("%d", &t);

    while (t > 0) {
        int n;
        scanf("%d", &n);
        printf("%d\n", findDigits(n));
```

Line: 32 Col: 2

QUESTION-3

[Problem Statement Link]=[Subtract the Product and Sum of Digits of an Integer - LeetCode]

[Source Code]

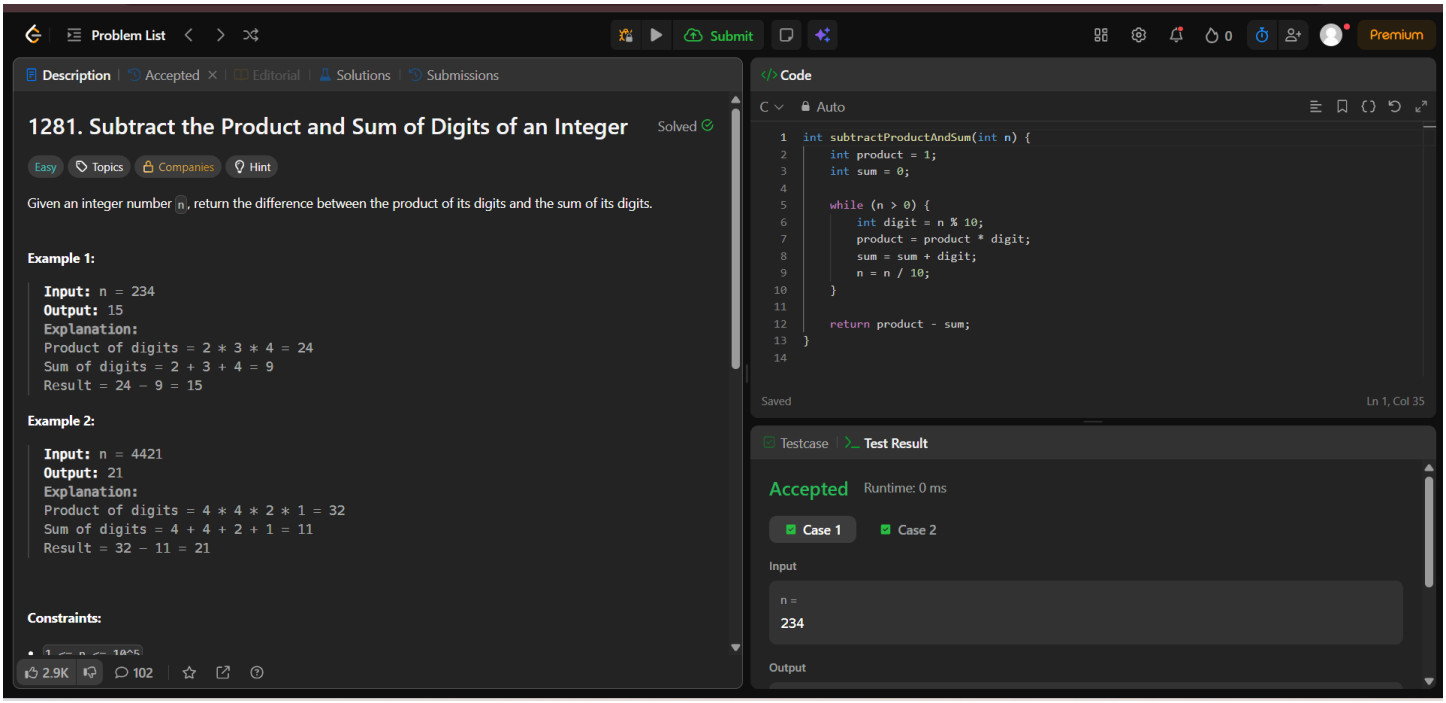
Code

C Auto

```c
int subtractProductAndSum(int n) {
    int product = 1;
    int sum = 0;

    while (n > 0) {
        int digit = n % 10;
        product = product * digit;
        sum = sum + digit;
        n = n / 10;
    }

    return product - sum;
}
```



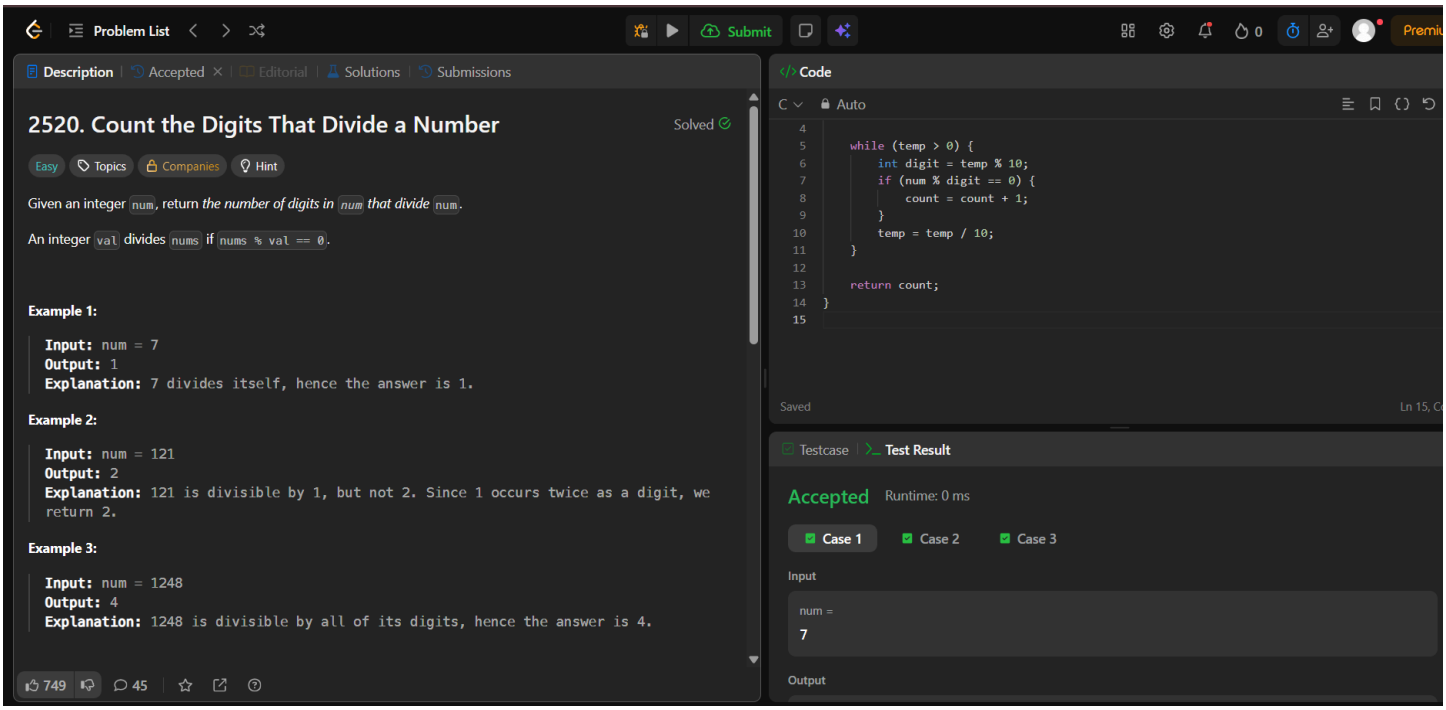
QUESTION-4

[Problem Statement Link]=[Count the Digits That Divide a Number - LeetCode]

[Source Code]

```c
    while (temp > 0) {
        int digit = temp % 10;
        if (num % digit == 0) {
            count = count + 1;
        }
        temp = temp / 10;
    }

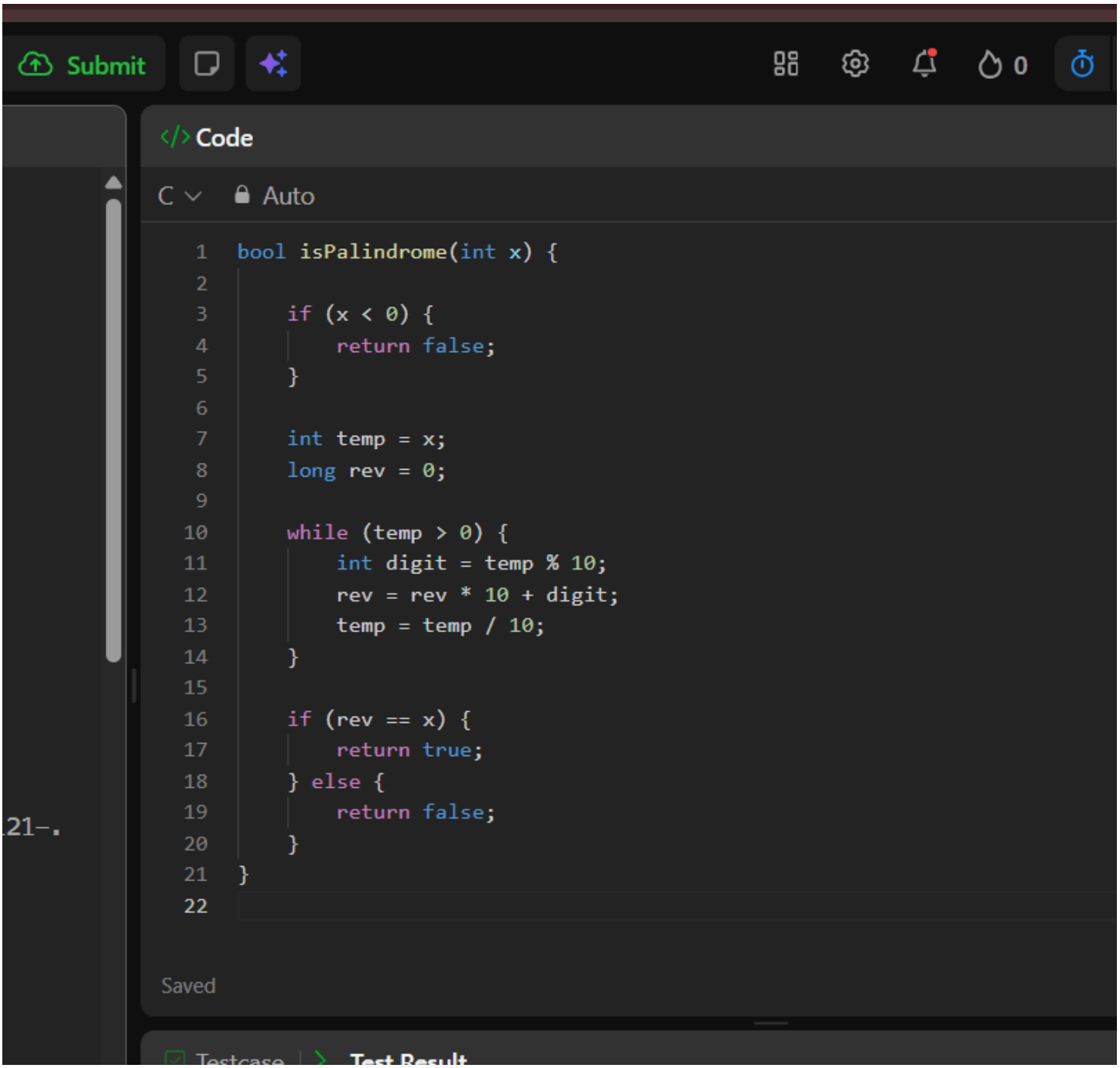
    return count;
}
```

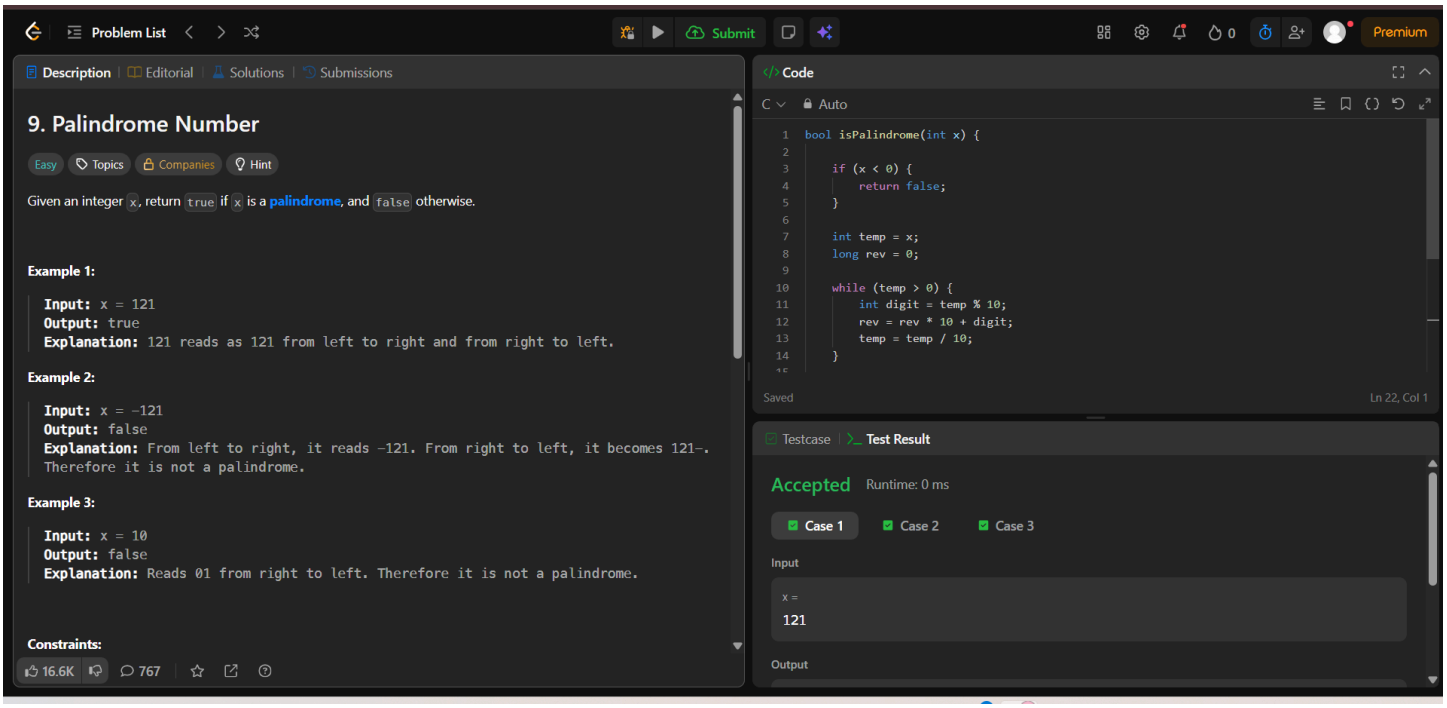


QUESTION–5

[Problem Statement Link]=[Palindrome Number - LeetCode]

[Source Code]


Submit
0
Code
C
Auto
bool isPalindrome(int x) {
if (x < 0) {
return false;
}
int temp = x;
long rev = 0;
while (temp > 0) {
int digit = temp % 10;
rev = rev * 10 + digit;
temp = temp / 10;
}
if (rev == x) {
return true;
} else {
return false;
}
}
Saved
Testcase
Test Result



QUESTION-6

[Problem Statement Link]=[Reverse Integer - LeetCode]

[Source Code]

Code

C Auto

```c
int reverse(int x){

    int rev = 0;

    while (x != 0) {
        int digit = x % 10;

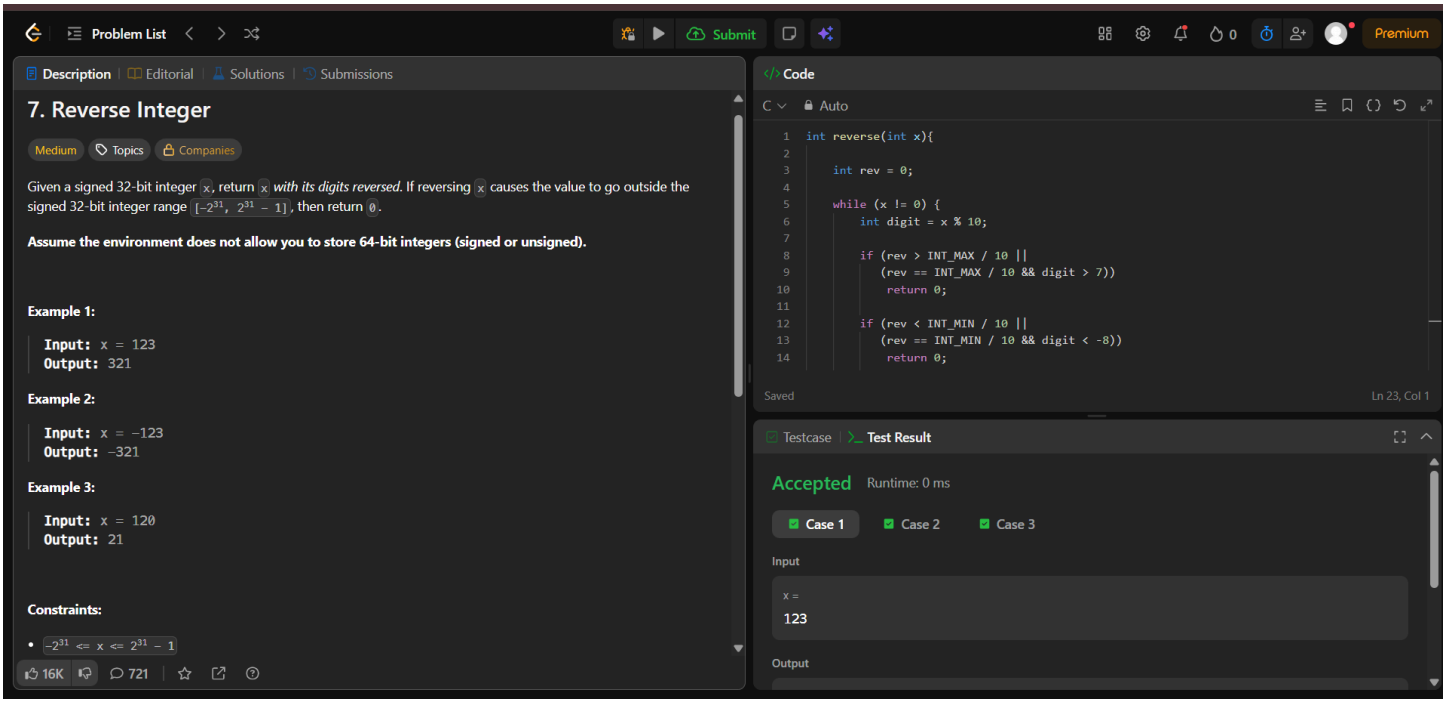
        if (rev > INT_MAX / 10 ||
           (rev == INT_MAX / 10 && digit > 7))
            return 0;

        if (rev < INT_MIN / 10 ||
           (rev == INT_MIN / 10 && digit < -8))
            return 0;

        rev = rev * 10 + digit;
        x = x / 10;
    }

    return rev;
}

```

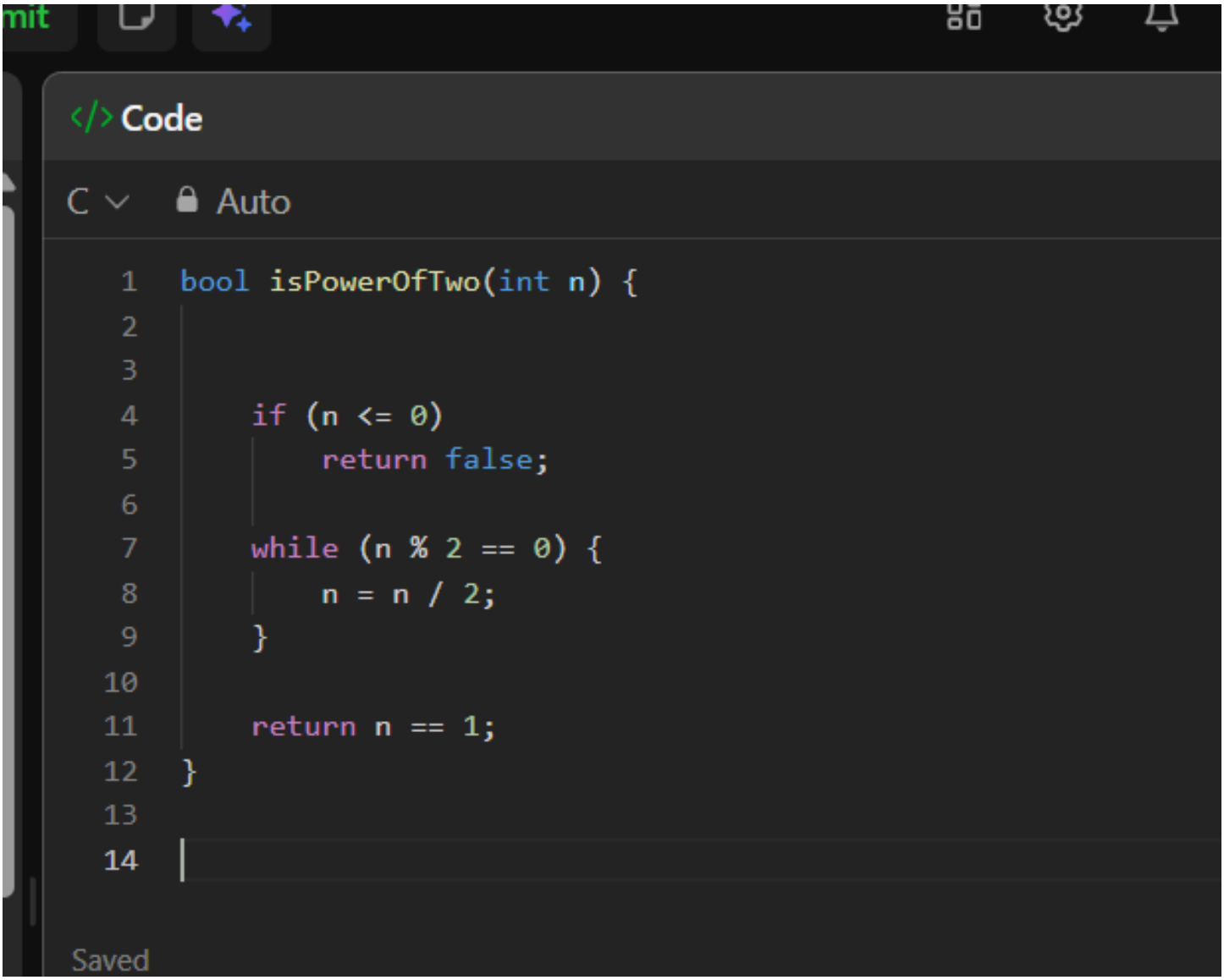


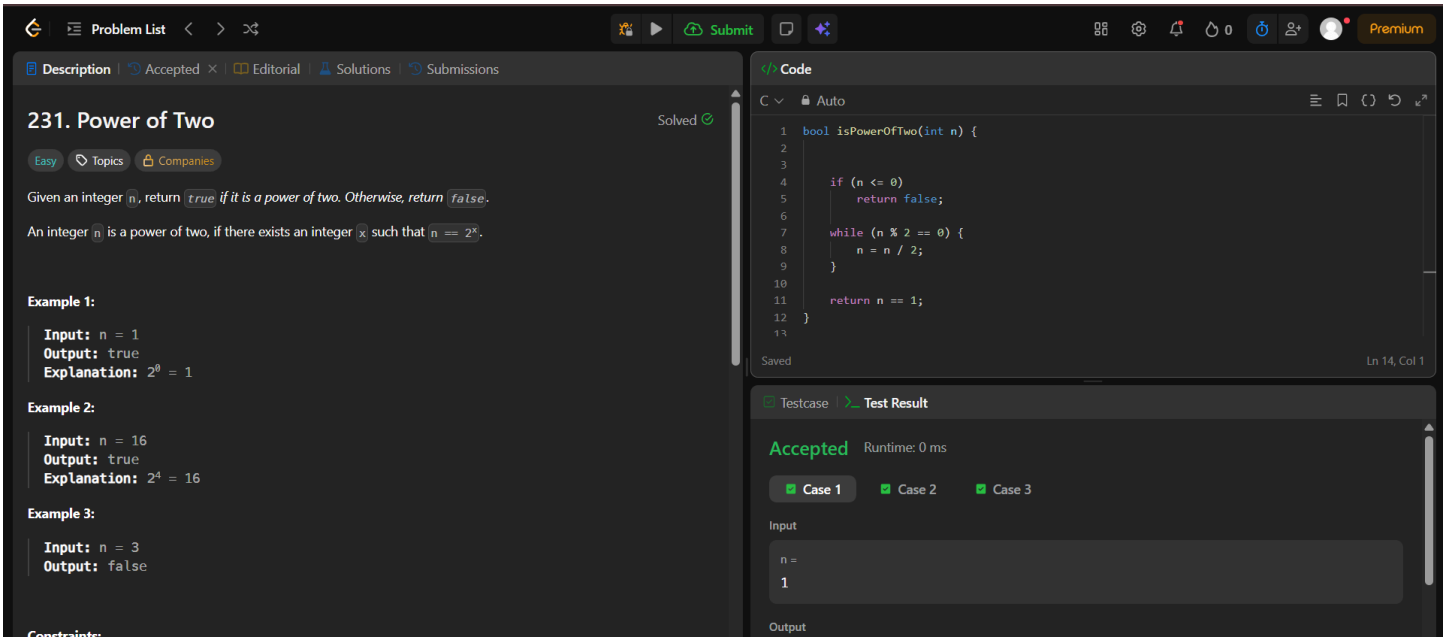
Module 5 — Factors, divisibility and mathematical loops

QUESTION-1

[Problem Statement Link]=[Power of Two - LeetCode]

[Source Code]

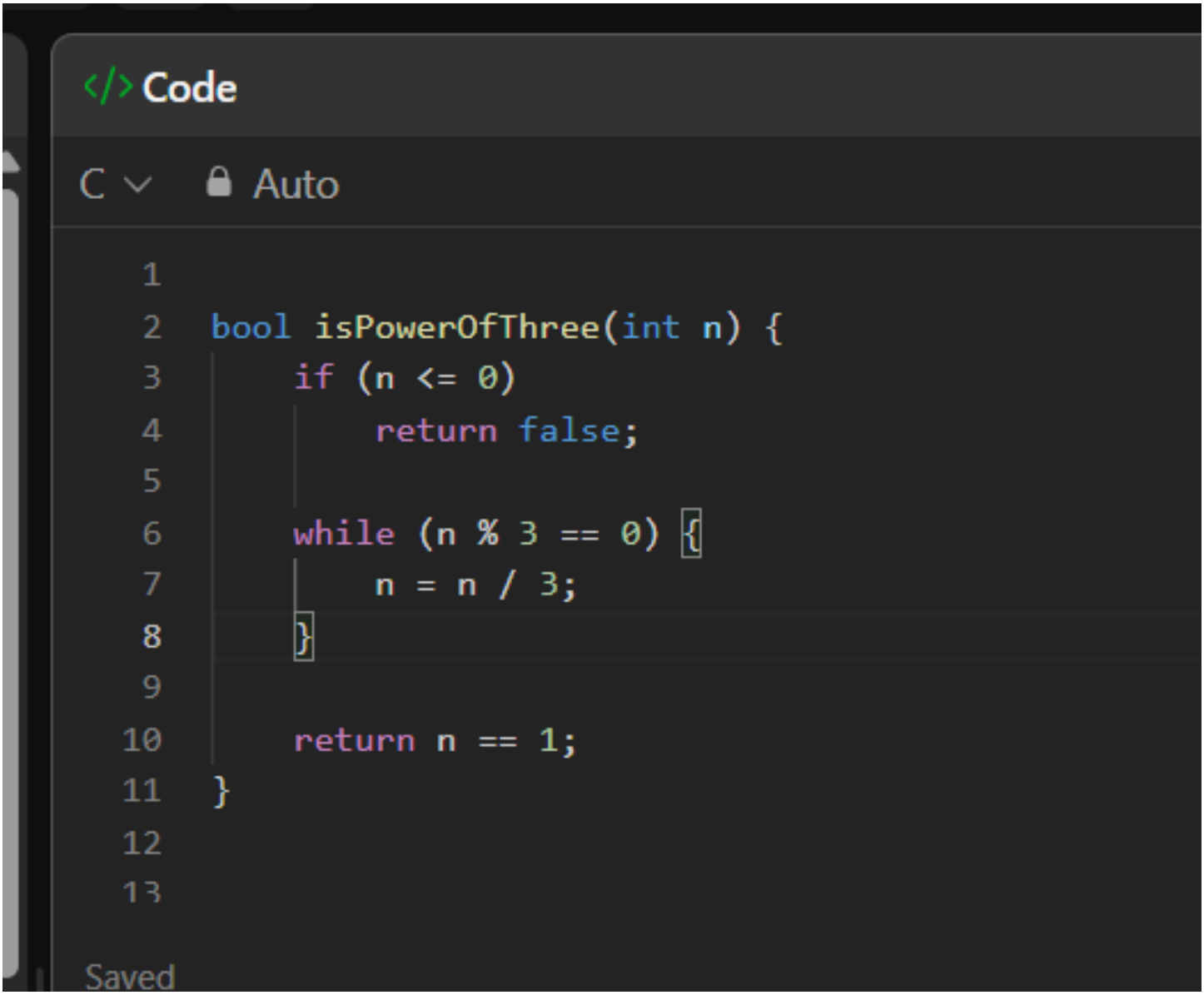

Code
C
Auto
bool isPowerOfTwo(int n) {
if (n <= 0)
return false;
while (n % 2 == 0) {
n = n / 2;
}
return n == 1;
}
Saved

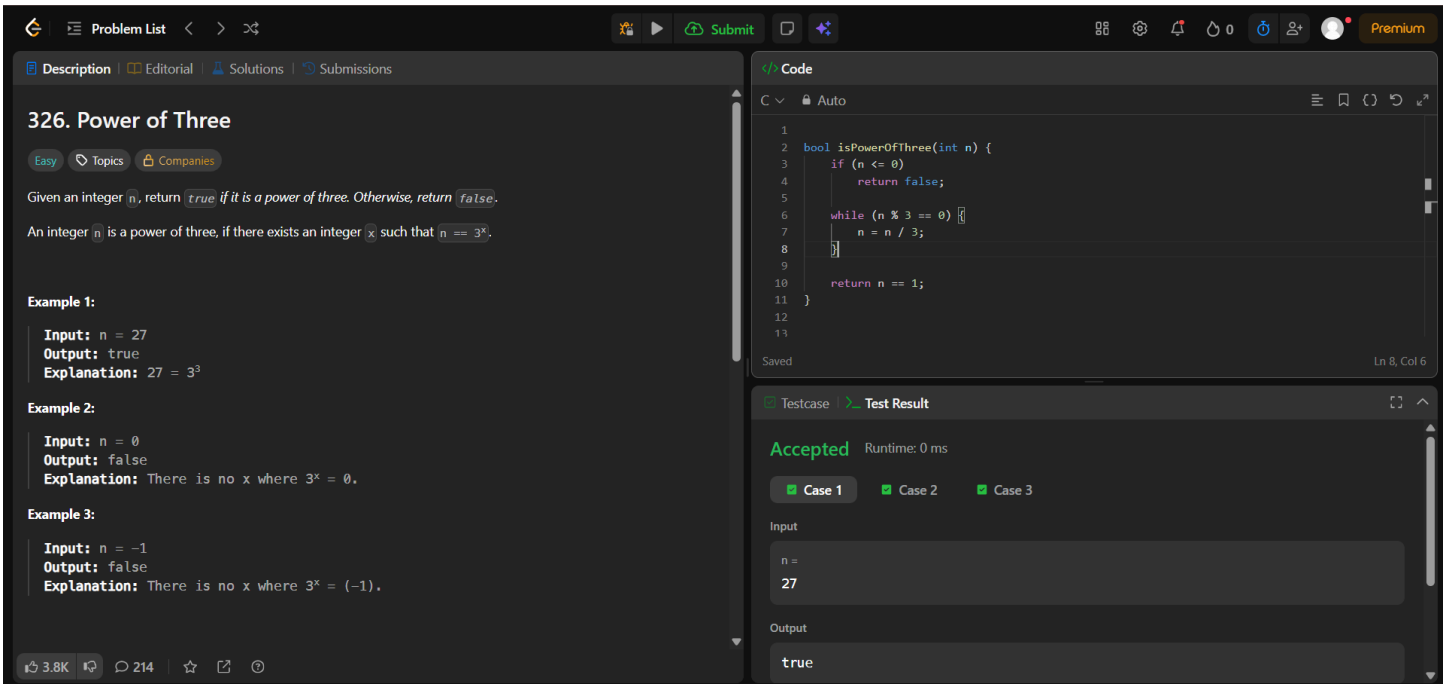

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
231. Power of Two
Solved
Easy
Topics
Companies
Given an integer n, return true if it is a power of two. Otherwise, return false.
An integer n is a power of two, if there exists an integer x such that n == 2^x.
Example 1:
Input: n = 1
Output: true
Explanation: 2^0 = 1
Example 2:
Input: n = 16
Output: true
Explanation: 2^4 = 16
Example 3:
Input: n = 3
Output: false
Code
C
Auto
bool isPowerOfTwo(int n) {
if (n <= 0)
return false;
while (n % 2 == 0) {
n = n / 2;
}
return n == 1;
}
Saved
Ln 14, Col 1
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Input
n =
1
Output

QUESTION–2

[Problem Statement Link]=[Power of Three - LeetCode]

[Source Code]

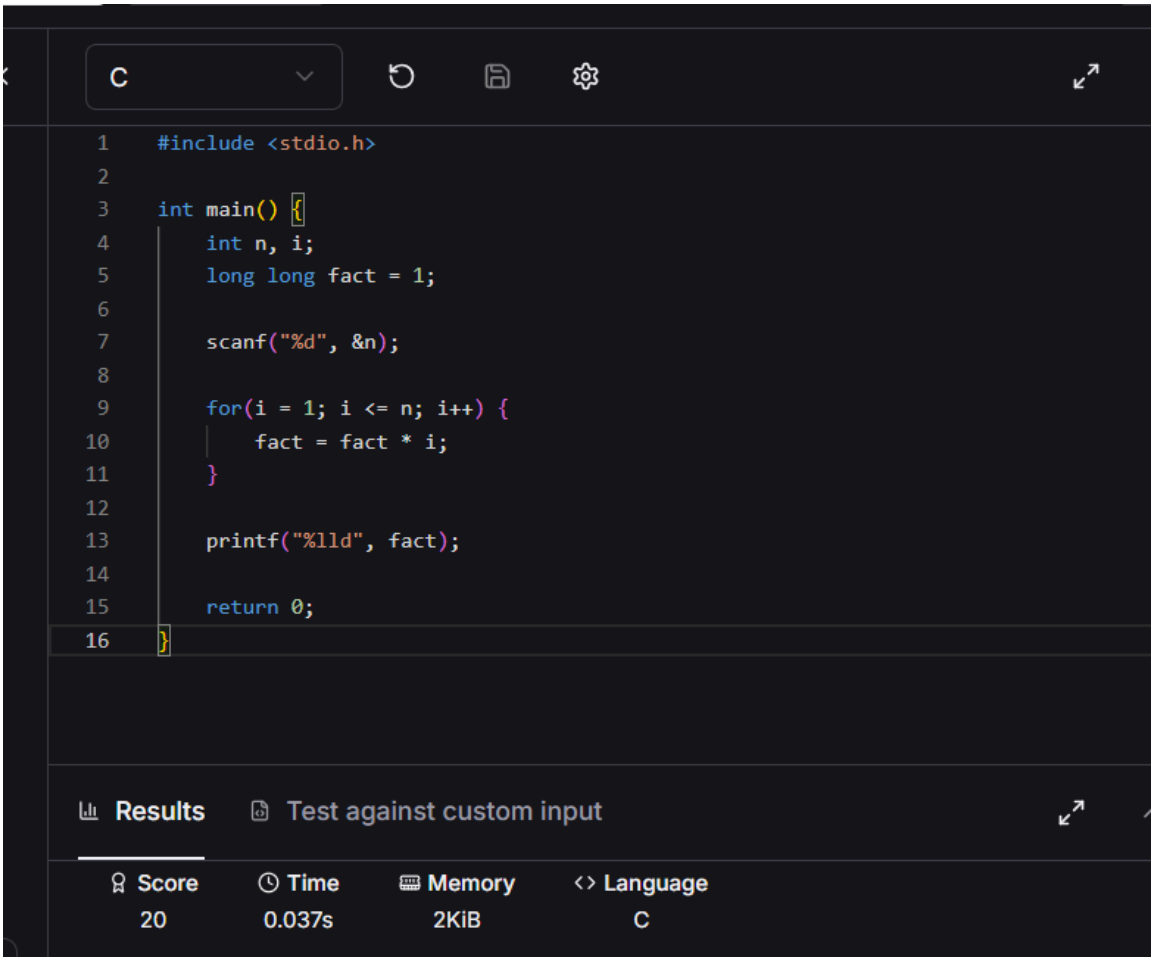




QUESTION–3

[Problem Statement Link]=[
🌐 Factorial! | Basics of Input/Output & Basic Programming Practice Problems]

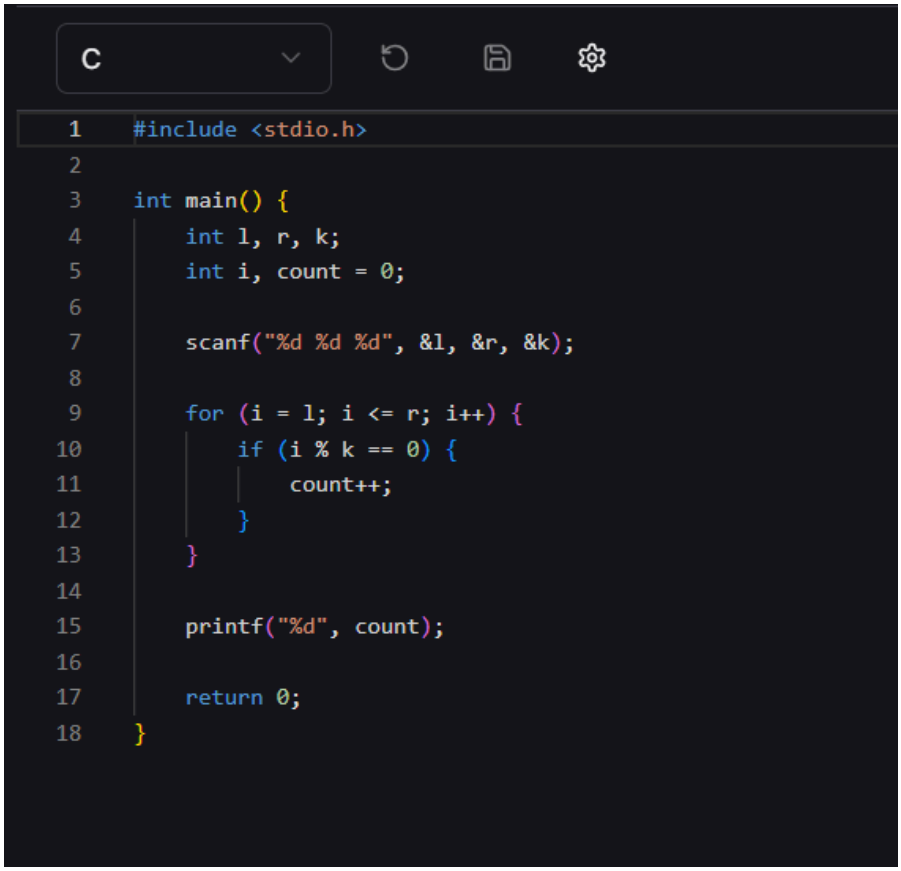
[Source Code]



QUESTION–4

[Problem Statement Link]=[
Count Divisors | Basics of Input/Output & Basic Programming Practice Problems]

[Source Code]



```c
#include <stdio.h>

int main() {
    int l, r, k;
    int i, count = 0;

    scanf("%d %d %d", &l, &r, &k);

    for (i = l; i <= r; i++) {
        if (i % k == 0) {
            count++;
        }
    }

    printf("%d", count);

    return 0;
}
```

QUESTION–5

[Problem Statement Link]=[Prime Numbers | Practice Problems]

[Source Code]

Enter your code or Upload your code as file.

```c
#include <stdio.h>

int main() {
    int n, i, count = 0;

    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        if(n % i == 0) {
            count++;
        }
    }

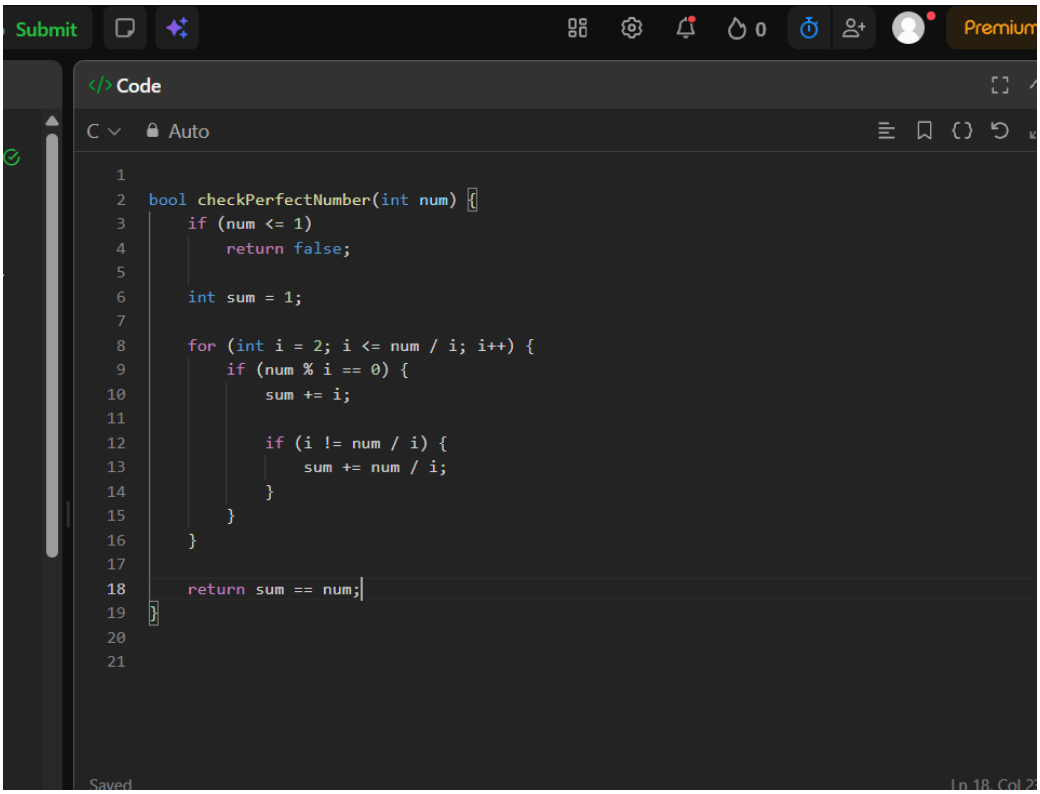
    if(count == 2)
        printf("1\n");
    else
        printf("0\n");

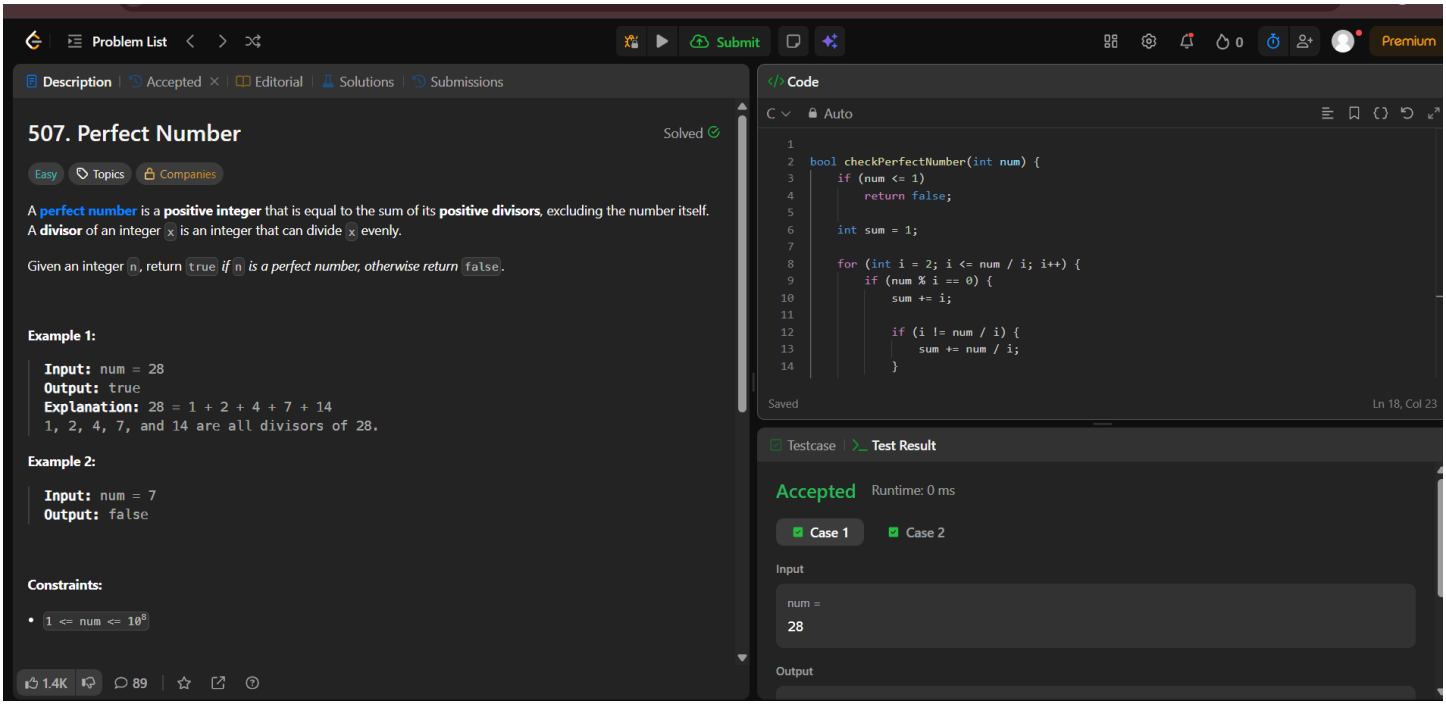
    return 0;
}
```

QUESTION-6

[Problem Statement Link]=[Perfect Number - LeetCode]

[Source Code]



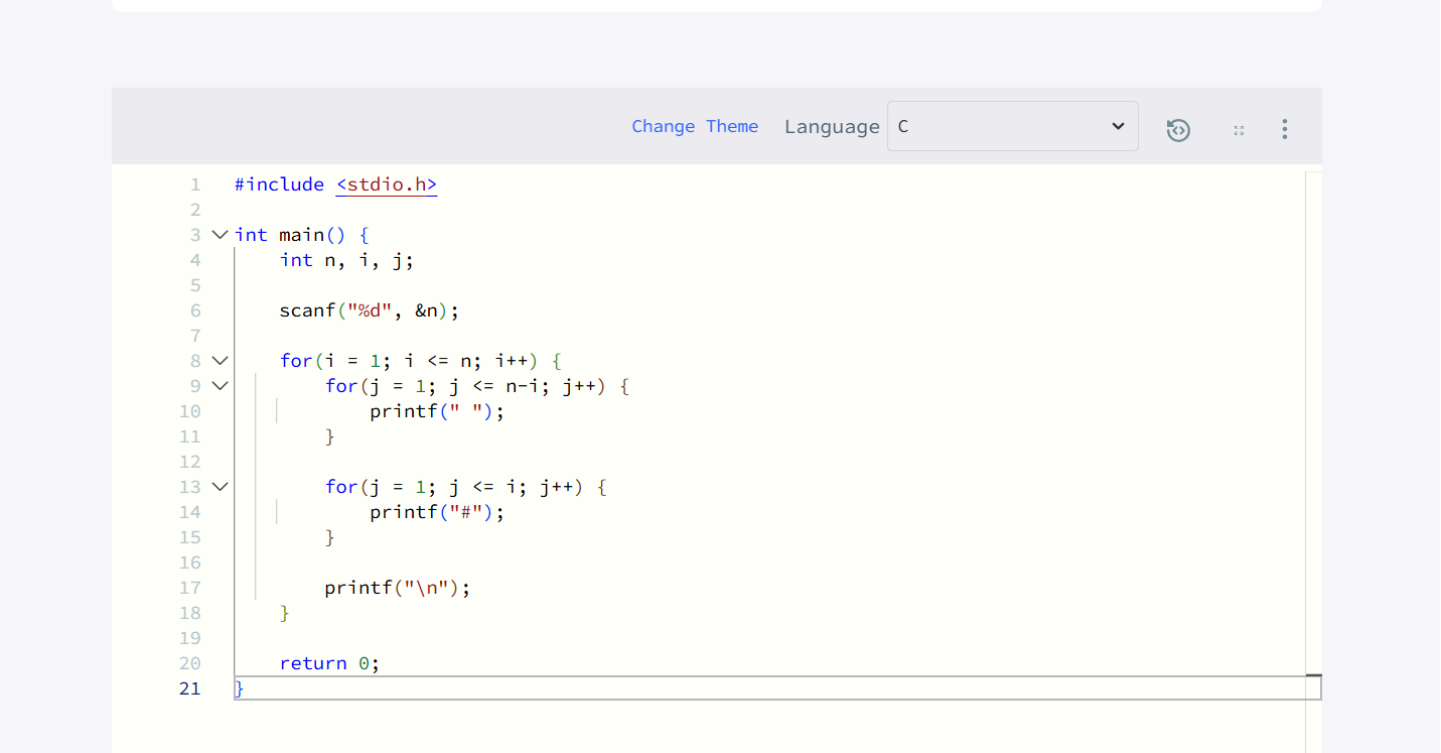


Module 6 — Nested loops: online judge problems

QUESTION–1

[Problem Statement Link]=[Staircase | HackerRank]

[Source Code]



```c
#include <stdio.h>

int main() {
    int n, i, j;

    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        for(j = 1; j <= n-i; j++) {
            printf(" ");
        }

        for(j = 1; j <= i; j++) {
            printf("#");
        }

        printf("\n");
    }

    return 0;
}
```

QUESTION–2

[Problem Statement Link]=[Printing Pattern Using Loops | HackerRank]

[Source Code]

Change Theme Language: C

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int size = 2 * n - 1;
    int i, j;

    for (i = 0; i < size; i++) {
        for (j = 0; j < size; j++) {
            int top = i;
            int bottom = size - 1 - i;
            int left = j;
            int right = size - 1 - j;

            int min = top;
            if (bottom < min) min = bottom;
            if (left < min) min = left;
            if (right < min) min = right;

            int value = n - min;
            printf("%d ", value);
        }
        printf("\n");
    }

```

Line: 27 Col: 1